

CRAIOVA, Romania

WMO: 154500

Lat: 44.3103N

Lon: 23.8669E

Elev: 193

StdP: 99.03

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.7	-14.3	1.1	-10.9	-11.6	1.4	-8.3	12.3	4.7	10.0	2.2	2.1	90	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	34.2	20.8	32.6	20.5	31.0	20.2	22.9	30.0	22.1	29.1	21.3	28.1	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	15.9	25.3	20.0	15.0	24.6	19.1	14.2	24.0	68.8	30.3	65.4	29.1	62.8	28.2	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.2	8.4	7.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-15.7	36.6	3.3	2.5	-18.1	38.4	-20.0	39.8	-21.9	41.2	-24.3	43.0		
			-15.9	24.9	3.3	1.5	-18.3	25.9	-20.2	26.8	-22.0	27.6	-24.4	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-0.8	2.1	7.0	12.4	17.4	21.3	23.5	23.4	18.4	12.1	6.3	0.6		
	DBStd	9.41	4.98	5.25	4.59	3.59	3.37	3.25	2.86	3.02	3.66	4.17	4.28	4.39		
	HDD10.0	1132	335	223	112	17	1	0	0	0	0	26	125	293		
	HDD18.3	2799	593	454	350	179	58	11	1	2	44	195	361	551		
	CDD10.0	1876	1	2	21	90	230	339	418	417	252	93	14	1		
	CDD18.3	500	0	0	0	2	29	99	161	161	46	3	0	0		
	CDH23.3	5099	0	0	1	34	284	990	1669	1661	424	37	0	0		
	CDH26.7	2005	0	0	0	5	49	359	724	737	127	4	0	0		
(14) Wind	WSAvg	3.0	2.6	3.2	3.6	3.7	3.2	2.8	2.8	2.7	2.9	2.8	2.9	2.7		
(15) Precipitation	PrecAvg	632	42	33	42	52	69	69	69	55	58	51	44	48		
	PrecMax	1121	96	64	130	130	149	138	187	215	134	136	108	143		
	PrecMin	305	5	3	7	0	11	13	9	2	2	1	1	0		
	PrecStd	177	24	19	30	27	34	34	47	48	40	36	30	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	17.3	21.9	27.1	30.1	35.0	37.4	36.7	32.6	26.9	20.0	13.3		
		MCWB	8.0	10.3	12.0	16.1	18.4	21.4	20.9	21.6	19.1	17.4	13.1	9.2		
	2%	DB	9.8	14.2	19.3	23.6	28.0	32.5	34.2	34.2	30.0	23.8	16.8	9.9		
		MCWB	6.6	8.8	11.3	15.1	18.1	20.7	20.9	20.8	18.4	16.2	12.6	7.4		
	5%	DB	7.4	11.6	17.0	21.2	26.5	30.4	32.3	32.5	27.9	21.4	14.2	7.7		
		MCWB	5.4	7.6	10.2	13.6	17.5	20.3	20.6	20.4	17.9	15.2	11.1	5.8		
	10%	DB	5.5	9.1	14.7	19.3	24.8	28.6	30.7	30.9	25.7	19.2	12.3	6.1		
		MCWB	4.1	6.3	8.9	12.6	16.8	19.7	20.4	20.1	17.1	14.2	10.2	4.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	11.2	13.5	17.2	20.8	23.4	24.1	24.3	21.4	19.0	14.2	9.8		
		MCDB	12.1	15.9	19.1	24.4	27.6	30.9	31.3	31.5	28.7	25.0	18.1	12.5		
	2%	WB	7.0	9.3	12.0	15.8	19.3	22.2	22.9	22.9	19.9	17.1	12.9	7.7		
		MCDB	9.1	13.3	17.5	22.2	25.7	29.5	29.8	30.2	26.3	21.3	16.0	9.5		
	5%	WB	5.7	8.0	10.7	14.6	18.3	21.3	22.1	21.9	18.8	15.8	11.7	6.2		
		MCDB	7.1	11.0	16.0	20.0	24.5	28.1	29.1	28.7	25.0	20.1	13.7	7.3		
	10%	WB	4.2	6.6	9.5	13.4	17.3	20.5	21.3	21.1	17.9	14.6	10.6	5.0		
		MCDB	5.5	8.8	14.1	18.0	23.1	26.8	28.1	27.9	23.9	18.5	12.1	5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.8	9.9	10.5	11.3	11.7	12.3	12.5	11.3	9.7	6.9	6.1		
		MCDBR	9.8	12.0	14.1	13.8	13.7	14.2	14.7	14.9	14.6	13.6	10.8	8.9		
		MCWBR	6.9	7.6	7.9	7.0	6.1	5.3	5.1	4.8	5.5	6.8	6.7	6.4		
		MCDBR	9.1	11.0	12.9	12.3	12.6	12.6	12.8	12.9	12.5	12.1	9.1	7.8		
	5% WB	MCWBR	6.6	7.2	7.7	6.6	6.3	5.5	5.3	5.3	5.8	6.6	5.9	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.338	0.367	0.403	0.460	0.438	0.439	0.452	0.470	0.433	0.393	0.366	0.340		
	taud		2.300	2.222	2.189	2.045	2.177	2.212	2.181	2.124	2.194	2.298	2.319	2.322		
	Ebn at Noon		762	803	823	803	836	835	818	785	783	766	721	718		
	Edh at Noon		84	108	127	160	144	140	143	146	126	100	83	75		
(44) All-Sky Solar Radiation	RadAvg		1.43	2.26	3.48	4.73	5.83	6.47	6.49	5.72	4.13	2.67	1.50	1.16		
	RadStd		0.22	0.35	0.40	0.47	0.44	0.47	0.41	0.43	0.34	0.36	0.21	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.56	-0.82	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		+0.55	N/A	N/A	N/A	N/A	N/A	N/A	-153	+79	N/A

Nomenclature: See separate page